



IRONTECH DOLL

BEAUTY INSPIRED CREATION

Irontech ULW White Paper 2026

Redefining Ownership Through Lightweight Engineering

**The Future Is Not Lighter.
The Future Is Easier.**

A branded overview of Irontech ULW Lightweight Technology, including design philosophy, verified production data, estimated model projections, testing principles, and practical user benefits.

Less Weight. More Comfort. Same Irontech Experience.

A Better Ownership Experience

For years, premium companion dolls have been defined by realism. As realism improved, products became larger, heavier, and more demanding to handle. Irontech ULW was developed to address that ownership challenge without moving away from the qualities collectors value most.

- Reduces overall body weight through an optimized lightweight internal structure.
- Focuses on easier lifting, repositioning, dressing, cleaning, storage, and everyday handling.
- Preserves the visual character, familiar touch, and structural confidence associated with Irontech bodies.
- Brings together four approved communication pillars: Up to 35% Lighter*, Realistic Look. Familiar Touch., Reliable Support. Lasting Durability., and Easier Everyday Use.

THE INDUSTRY PROBLEM

When Realism Becomes Harder to Live With

Heavier products can look impressive, but ownership becomes more physically demanding over time. For many collectors, the challenge is no longer choosing the right body. It is handling it comfortably in everyday life.

- Lifting a full-size body onto a bed or sofa.
- Repositioning a body repeatedly during setup or use.
- Changing outfits, cleaning, and long photography sessions.
- Managing larger and fuller body styles without excessive physical strain.

ULW was created to make ownership easier—not by making the body less realistic, but by making it smarter inside.

CORE VALUE

Lighter, Without Losing What Matters

ULW is built around a simple product principle: reduce weight while preserving the experience users chose the body for in the first place.

- Up to 35% Lighter* — a brand-level communication claim for selected ULW models and configurations.
- Realistic Look. Familiar Touch. — designed to preserve body contours, visual fullness, and the familiar silicone response.
- Reliable Support. Lasting Durability. — engineered for structural confidence and repeated daily handling.
- Easier Everyday Use — less effort across lifting, repositioning, dressing, cleaning, posing, and storage.

PRECISION MANUFACTURING

Engineering Beyond Weight Reduction

ULW is not a simple material reduction exercise. It reflects a more controlled production approach intended to improve internal efficiency while respecting the external experience.

- Traditional direction: faster pouring with slower curing.
- ULW direction: ultra-slow pouring at approximately 3–5 grams per second with faster curing control.
- Five independent pouring zones with customized speed and process parameters.
- Real-time monitoring by two technicians during production.
- Three industrial endoscopes used to help observe air pockets, voids, and structural weak points.
- A single ULW body typically requires 1–2 hours of focused production time, compared with only minutes for standard bodies.

ULW is not about doing less. It is about doing more—with greater precision and care.

VERIFIED IRONTECH DATA

Weight Reduction Data

The following figures reflect Irontech-only actual production records. Records beginning with LN or RL are not included in this section. Within the current verified Irontech dataset, ULW has been documented across 20 production records covering 7 body models.

20	7	4.9 kg	14.3%	22.6%
Production Records	Body Models	Average Weight Saved	Average Weight Reduction	Highest Verified Reduction

Selected Verified Results

160 22.6% Lighter 44.6 kg → 34.5 kg Standard ULW 10.1 kg Weight Saved	158T 19.9% Lighter 34.7 kg → 27.8 kg Standard ULW 6.9 kg Weight Saved	154 18.8% Lighter 33.0 kg → 26.8 kg Standard ULW 6.2 kg Weight Saved	164 17.9% Lighter 43.0 kg → 35.3 kg Standard ULW 7.7 kg Weight Saved
--	--	---	---

These records show that larger and fuller Irontech bodies often deliver more meaningful absolute weight savings, making the difference especially noticeable in lifting and repositioning.

PROJECTED RESULTS

Estimated Performance for Selected Models

Some Irontech models are currently represented by projected figures rather than final confirmed production data. These estimates should be presented separately from verified results.

PR190 Estimated 31.3% Lighter	PR178 Estimated 25.8% Lighter	162P Estimated 24.7% Lighter
---	---	--

≈80 kg → ≈55 kg Standard Estimated ULW ≈25 kg Estimated Saving	≈48.5 kg → ≈36.0 kg Standard Estimated ULW ≈12.5 kg Estimated Saving	≈55.8 kg → ≈42.0 kg Standard Estimated ULW ≈13.8 kg Estimated Saving
---	---	---

Projected figures are estimates only and must be confirmed through actual production and weighing.

VALIDATION & TESTING

Tested for the Real World

ULW development and evaluation extend beyond the scale. The objective is to verify that lighter weight does not come at the cost of touch, stability, or durability.

- Touch and softness evaluation.
- Waist twisting, arm lifting, and torso folding.
- Knee articulation, sitting, kneeling, and full-split positions.
- Long-term deformation resistance and structural stability observation.
- Extreme position tests, including full split for over 2 hours, sitting and kneeling positions maintained for 12 consecutive hours each, and a continuous 9-day durability and stability test.

Observed outcomes: no tearing, no abnormal bulging, no structural failures. In internal evaluation, multiple testers also reported that some ULW bodies felt softer than their standard counterparts.

USER BENEFIT

Who Benefits Most From ULW

The practical value of ULW becomes clearer in everyday ownership. The technology is especially relevant for users who want a realistic full-size experience with less physical burden.

- First-time owners seeking a more approachable experience.
- Collectors of fuller body types who do not want weight to limit their choice.
- Mature collectors seeking less strain during lifting and positioning.
- Female owners and other users who value easier handling and accessibility.
- Photographers and content creators who need faster setup and more frequent repositioning.

CONCLUSION

The Next Standard of Ownership

ULW is more than a weight-reduction feature. It reflects Irontech's broader belief that premium ownership should be more comfortable, more practical, and more enjoyable over time.

Less Weight. More Comfort. Same Irontech Experience.

Irontech measures innovation by the value it creates for collectors. ULW is one expression of that commitment.

DISCLAIMER

Important Notes on Data Usage

This white paper combines verified production data and projected model estimates. These two categories should not be treated as identical evidence levels.

- *Up to 35% lighter applies to selected ULW models and configurations as an approved brand communication claim.
- Within the current Irontech-only verified production records presented here, the highest confirmed reduction is approximately 22.6%.
- Projected results are estimates and must be confirmed by actual production and weighing before external use as verified data.
- Actual product weight and weight-reduction percentage vary according to body size, internal structure, functional configuration, softness options, and customization.